



Two-step equations

These need two different opposite operations to solve them.

For simple two-step equations, usually the coefficient $\neq 1$.



Remember that what is done to one side must be done to the other.

Solve these equations by isolating the variable using inverse operations.

(i) $2x + 3 = 7$



Coefficient of $x = 2$

$$2x + 3 - 3 = 7 - 3$$

$$2x = 4$$

$$2x \div 2 = 4 \div 2$$

$$\therefore x = 2$$

Subtract 3 from both sides

x is still not by itself

Divide both sides by 2

(ii) $5 + 2a = 20$

$$5 - 5 + 2a = 20 - 5$$

$$2a = 15$$

$$2a \div 2 = 15 \div 2$$

$$\therefore a = 7\frac{1}{2}$$

Subtract 5 from both sides

a is still not by itself

Divide both sides by 2

(iii) $10 - d = 22$

$$10 - 10 - d = 22 - 10$$

$$-d = 12$$

$$-d \div (-1) = 12 \div (-1)$$

$$\therefore d = -12$$



Coefficient of d is -1

Subtract 10 from both sides

d is still not by itself

Divide both sides by (-1)

Positive divided by a negative gives a negative answer

(iv) $-4k + 14 = 3$

$$-4k + 14 - 14 = 3 - 14$$

$$-4k = -11$$

$$-4k \div (-4) = -11 \div (-4)$$

$$\therefore k = 2\frac{3}{4} \text{ or } 2.75$$

Subtract 14 from both sides

k is still not by itself

Divide both sides by (-4)

Take care with negative signs





Two-step equations



1 Use inverse operations to solve these two-step equations.

a $3x + 5 = 14$

$$\begin{array}{l}
 3x + 5 - 5 = 14 - 5 \\
 3x = \boxed{} \\
 3x \div \boxed{} = \boxed{} \div \boxed{} \\
 x = \boxed{}
 \end{array}$$

b $6g - 1 = 27$

$$\begin{array}{l}
 6g - 1 + 1 = 27 + 1 \\
 6g = \boxed{} \\
 6g \div \boxed{} = \boxed{} \div \boxed{} \\
 g = \boxed{}
 \end{array}$$

c $4a + 11 = 39$

$$\therefore a = \boxed{}$$

d $5w - 12 = 18$

$$\therefore w = \boxed{}$$

e $2b + 12 = 8$

$$\therefore b = \boxed{}$$

f $3n - 5 = -8$

$$\therefore n = \boxed{}$$

g $10 - 2m = 18$

$$\begin{array}{l}
 10 - 2m - 10 = 18 - 10 \\
 -2m = \boxed{} \\
 -2m \div \boxed{} = \boxed{} \div \boxed{} \\
 m = \boxed{}
 \end{array}$$

h $25 - 6p = -29$

$$\therefore p = \boxed{}$$

i $3 - g = 8$

$$\therefore g = \boxed{}$$

j $15 - y = 16$

$$\therefore y = \boxed{}$$

k $9 - q = 4$

$$\therefore q = \boxed{}$$

l $25 = 13 - n$

$$\therefore n = \boxed{}$$

Equations with variables on both sides

The aim is to get all the variables on one side and the numbers on the other side of the equation.

Solve these equations using inverse operations.

(i) $5x = x + 8$

$$5x - x = x - x + 8$$

Subtract x from both sides

$$4x = 8$$

Numbers and variable on different sides

$$4x \div 4 = 8 \div 4$$

Get the variable x by itself

$$\therefore x = 2$$

(ii) $6b = 9b - 12$

$$6b - 9b = 9b - 9b - 12$$

Subtract $9b$ from both sides

$$-3b = -12$$

Numbers and variable on different sides

$$-3b \div (-3) = -12 \div (-3)$$

Get the variable b by itself

$$\therefore b = 4$$

(iii) $3h - 35 = 4 - 5h$

$$3h + 5h - 35 = 4 - 5h + 5h$$

Add $5h$ to both sides

$$8h - 35 = 4$$

$$8h - 35 + 35 = 4 + 35$$

Add 35 to both sides

$$8h = 39$$

Numbers and variable on different sides

$$8h \div 8 = 39 \div 8$$

Get the variable h by itself

$$\therefore h = 4\frac{7}{8} \text{ or } 4.875$$

(iv) $2p + 18 - 8p = 24 - 3p$

$$18 - 6p = 24 - 3p$$

Simplify by collecting like terms

$$18 - 6p + 6p = 24 - 3p + 6p$$

Add $6p$ to both sides

$$18 = 24 + 3p$$

$$18 - 24 = 24 - 24 + 3p$$

Subtract 24 from both sides

$$-6 = 3p$$

Get the variable p by itself

$$-6 \div 3 = 3p \div 3$$

$$-2 = p \text{ or } p = -2$$



Equations with variables on both sides



- 1 Use inverse operations to solve these equations containing variables on both sides.

a $10b = 5b + 20$

b $18 - 4x = 2x$

$\therefore b = \boxed{}$

$\therefore x = \boxed{}$

c $3d = 18 - 5d$

d $w + 20 = 7w - 4$

$\therefore d = \boxed{}$

$\therefore w = \boxed{}$

e $g + 13 = 2g - 5$

f $5m - 7 = 2m - 1$

$\therefore g = \boxed{}$

$\therefore m = \boxed{}$

g $2t - 12 = 12 + 6t$

h $16 - 2k = 9 - 5k$

$\therefore t = \boxed{}$

$\therefore k = \boxed{}$

**Equations with variables on both sides**

2 Use inverse operations to solve these equations containing variables on both sides.

a $17 - 2k = k + 5$

$\therefore k = \boxed{}$

b $8 + 2g = 8 - 2g$

$\therefore g = \boxed{}$

c $-x - 11 = x - 19$

$\therefore x = \boxed{}$

d $-16 + 9w = 4 - w$

$\therefore w = \boxed{}$

e $2y - 21 + 6y = 18 + 5y$

$\therefore y = \boxed{}$

f $5a + 4a + 9 = 21a - 39$

$\therefore a = \boxed{}$

g $10 + 2r - 3r + 4 = r - 8$

$\therefore r = \boxed{}$

h $5p - 13 = 3p - 21 + 4p$

$\therefore p = \boxed{}$

Equations with fractions

For equations with a fraction, multiply both sides by the value in the denominator.

Use inverse operations to solve these equations containing fractions.

(i) $\frac{5a}{4} = -15$



$$\frac{5a}{4} \times 4 = -15 \times 4$$

Multiply both sides by the denominator (4)

$$5a = -60$$

a is still not by itself

$$5a \div 5 = -60 \div 5$$

Divide both sides by 5

$$\therefore a = -12$$



$\frac{5a}{4} = \frac{5}{4} \times a$, so another way to solve this simple equation is to divide both sides by $\frac{5}{4}$

(ii) $\frac{m-6}{2} = 4$

$$\frac{m-6}{2} \times 2 = 4 \times 2$$

Multiply both sides by the denominator (2)

$$m-6 = 8$$

m is still not by itself

$$m-6+6 = 8+6$$

Add 6 to both sides

$$\therefore m = 14$$

(iii) $-5 + \frac{y}{6} = 3$



When only the variable term is in fraction form, we remove the other terms first.

$$-5 + 5 + \frac{y}{6} = 3 + 5$$

Add 5 to both sides

$$\frac{y}{6} = 8$$

y is still not by itself

$$\frac{y}{6} \times 6 = 8 \times 6$$

Multiply both sides by the denominator (6)

$$\therefore y = 48$$

(iv) $\frac{n}{4} + 9 = -2\frac{1}{2}$

$$\frac{n}{4} + 9 - 9 = -2\frac{1}{2} - 9$$

Subtract 9 from both sides

$$\frac{n}{4} = -11\frac{1}{2}$$

n is still not by itself

$$\frac{n}{4} \times 4 = -11\frac{1}{2} \times 4$$

Multiply both sides by the denominator (4)

$$\therefore n = -46$$

Positive $\times$ negative = negative



Equations with fractions

Solve each of the equations below containing fractions.

1 a $\frac{3v}{4} = 6$

$$\frac{3v}{4} \times 4 = 6 \times 4$$

$$3v = \boxed{}$$

$$3v \div \boxed{} = \boxed{} \div \boxed{}$$

$$\therefore v = \boxed{}$$

b $\frac{7t}{3} = 21$

$$\therefore t = \boxed{}$$

c $\frac{3u}{2} = -12$

$$\therefore u = \boxed{}$$

d $\frac{-6h}{5} = 4$

$$\therefore h = \boxed{}$$

2 a $\frac{x+4}{3} = 6$

$$\frac{x+4}{3} \times 3 = 6 \times 3$$

$$x+4 = \boxed{}$$

$$x+4 \boxed{} = \boxed{} - \boxed{}$$

$$\therefore x = \boxed{}$$

b $\frac{y-12}{4} = 2$

$$\therefore y = \boxed{}$$

c $\frac{d-16}{2} = -9\frac{1}{2}$

$$\therefore d = \boxed{}$$

d $-5 = \frac{8+a}{2}$

$$\therefore a = \boxed{}$$



Equations with fractions

Solve each of the equations below containing fractions.

2 e $12 = \frac{4+b}{-3}$

$\therefore b = \boxed{}$

f $\frac{-0.5+p}{3} = -6$

$\therefore p = \boxed{}$

3 a $\frac{y}{3} + 2 = 5$

$$\begin{aligned}\frac{y}{3} + 2 - 2 &= 5 - 2 \\ \frac{y}{3} &= \boxed{} \\ \frac{y}{3} \times \boxed{} &= \boxed{} \times \boxed{} \\ \therefore y &= \boxed{}\end{aligned}$$

b $\frac{k}{5} - 10 = 2$

$\therefore k = \boxed{}$

c $4 + \frac{h}{5} = 6$

$\therefore h = \boxed{}$

d $\frac{m}{7} + 5 = 1$

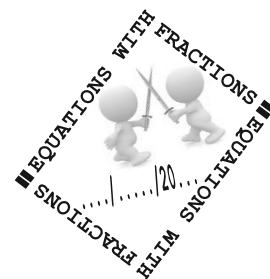
$\therefore m = \boxed{}$

e $-3 + \frac{x}{5} = -6$

$\therefore x = \boxed{}$

f $\frac{t}{6} + 1\frac{1}{2} = 5$

$\therefore t = \boxed{}$



Equations with parentheses

This first method works best for simple equations with only one pair of parentheses.

Solve these equations containing parentheses.

(i) $2(x + 2) = 6$

$$2(x + 2) \div 2 = 6 \div 2$$

$$x + 2 = 3$$

$$x + 2 - 2 = 3 - 2$$

$$\therefore x = 1$$

Divide both sides by 2

Get the variable x all by itself

(ii) $-(2h - 5) = 4$

$$-(2h - 5) \div (-1) = 4 \div (-1)$$

$$2h - 5 = -4$$

$$2h - 5 + 5 = -4 + 5$$

$$2h = 1$$

$$2h \div 2 = 1 \div 2$$

$$\therefore h = \frac{1}{2}$$

Remove the $\times(-1)$

Get the variable h all by itself

means $\times(-1)$



Remember: Distributive law:

$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac$$

You can use the distributive law to expand the parentheses first.



Solve these equations after first expanding using the distributive law.

(i) $3(a - 7) = 4a$

$$3(a - 7) = 4a$$

$$3a - 21 = 4a$$

$$3a - 3a - 21 = 4a - 3a$$

$$-21 = a$$

$$\therefore a = -21$$

Expand the LHS

Subtract $3a$ from both sides

(ii) $7 = -6(d + 4)$

$$7 = -6(d + 4)$$

$$7 = -6d - 24$$

$$7 + 24 = -6d - 24 + 24$$

$$31 = -6d$$

$$31 \div (-6) = -6d \div (-6)$$

$$\therefore -5\frac{1}{6} = d$$

Expand the RHS

Get the variable d all by itself



Equations with parentheses



1 Solve these equations without using the distributive law:

a $2(a + 4) = 12$

b $7(n - 3) = 14$

$$2(a + 4) \div 2 = 12 \div 2$$

$$\begin{aligned} & \boxed{} = 6 \\ & \boxed{} - \boxed{} = 6 - \boxed{} \\ & \therefore a = \boxed{} \end{aligned}$$

$$\therefore n = \boxed{}$$

c $-(d + 3) = 11$

d $3(2y + 1) = 18$

$$\therefore d = \boxed{}$$

$$\therefore y = \boxed{}$$

e $-(4r + 1) = 19$

f $\frac{1}{2}(x + 3) = -2$

$$\therefore r = \boxed{}$$

$$\therefore x = \boxed{}$$

g $-2(6 - k) = 5$

h $4(5 - 2w) = 4\frac{1}{2}$

$$\therefore k = \boxed{}$$

$$\therefore w = \boxed{}$$



Equations with parentheses

2 Solve these equations after first expanding using the distributive law:

a $2(b-3) = 8$

$$\begin{array}{l}
 \overset{\times}{\curvearrowright} \\
 2(b-3) = 8 \\
 2b-6 = 8 \\
 2b-6 + \boxed{} = 8 + \boxed{} \\
 2b \div \boxed{} = 14 \div \boxed{} \\
 b = \boxed{}
 \end{array}$$

b $3(w+1) = 18$

$\therefore w = \boxed{}$

c $-(q+2) = 3$

$\therefore q = \boxed{}$

d $-(x-6) = -10$

$\therefore x = \boxed{}$

e $-5(t+2) = 25$

$\therefore t = \boxed{}$

f $2(3m+5) = 1$

$\therefore m = \boxed{}$

g $-2(8+5p) = -24$

$\therefore p = \boxed{}$

h $\frac{1}{2}(4a+8) = -9$

$\therefore a = \boxed{}$

Combo time: Multi-step equations

These require at least two or more steps to solve.

Solve these equations by isolating the variable using inverse operations.

(i) $\frac{6+11d}{4} = 7$

$$\frac{6+11d}{4} \times 4 = 7 \times 4$$

Multiply both side by 4

$$6 + 11d = 28$$

d is still not by itself

$$6 + 11d - 6 = 28 - 6$$

Subtract 6 from both sides

$$11d = 22$$

d is still not by itself

$$11d \div 11 = 22 \div 11$$

Divide both sides by 11

$$\therefore d = 2$$

(ii) $\frac{4x+10}{4} = \frac{x}{6}$



Multiply each numerator by the opposite denominator if both sides are fractions.

$$\frac{4x+10}{4} = \frac{x}{6}$$

"Cross multiply" the denominators

$$6 \times (4x + 10) = 4 \times x$$

$$6(4x + 10) = 4x$$

Expand the left-hand side

$$24x + 60 = 4x$$

$$24x - 24x + 60 = 4x - 24x$$

Subtract $24x$ from both sides

$$60 = -20x$$

$$60 \div (-20) = -20x \div (-20)$$

Divide both sides by -20

$$\therefore -3 = x$$

(iii) $3(2 - 3m) = 5(m - 6)$

$$3(2 - 3m) = 5(m - 6)$$

Expand both sides

$$6 - 9m = 5m - 30$$

$$6 - 9m - 5m = 5m - 5m - 30$$

Subtract $5m$ from both sides

$$6 - 14m = -30$$

$$6 - 6 - 14m = -30 - 6$$

Subtract 6 from both sides

$$-14m = -36$$

$$-14m \div (-14) = -36 \div (-14)$$

Divide both sides by -14

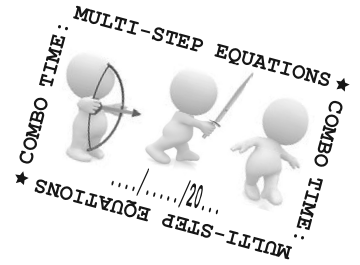
$$\therefore m = 2\frac{4}{7}$$

**Combo time: Multi-step equations**

Use opposite operations to solve these multi-step equations.

$$1 \quad \frac{2x-5}{3} = -9$$

$$2 \quad \frac{7+5y}{2} = 16$$



$$\therefore x = \boxed{}$$

$$\therefore y = \boxed{}$$

$$3 \quad \frac{5b+1}{7} = \frac{2b}{3}$$

$$4 \quad \frac{3a-2}{5} = \frac{4a}{6}$$

$$\therefore b = \boxed{}$$

$$\therefore a = \boxed{}$$

**Combo time: Multi-step equations**

$$5 \quad \frac{15h}{4} = \frac{14h+11}{3}$$

$$6 \quad 3(2+q) = 6(4+q)$$

$$\therefore h = \boxed{}$$

$$\therefore q = \boxed{}$$

$$7 \quad 3(4n+1) = 5(4-n)$$

$$8 \quad -8(k-2) = 6(k+2)$$

$$\therefore n = \boxed{}$$

$$\therefore k = \boxed{}$$



Equations challenge!

How many different equations can you make that give the answers below?

Scoring:

- 1-step equations = 1 point
- With variables on both sides = 3 points
- 2-step equations = 2 points
- 3 or more step equations = 5 points



$$2(5a + 12) = 44$$

= 5 points!

$$a = 2$$

$$n = 5$$

$$x = 9$$



$$6x - 81 = -3x$$

= 3 points!



Word problems

Word problems can be changed into equations to help solve.

Write matching algebraic equations for these statements:

Unless you are given a variable to use, you can choose any letter to represent a number.

- (i) A number added to 12 equals 21.

Let the number be n

Pick a letter to represent the number

$$\therefore 12 + n = 21$$

Write the equation using the chosen variable

- (ii) 19 subtracted from double a number leaves 23.

Let the number be x

Pick a letter to represent the number

$$\therefore 2x - 19 = 23$$

Write the equation using the chosen variables



The order that the numbers in a subtraction or division are written is important.

- (iii) Jennifer walks d metres to the shops to meet her friends. Kylie walks an extra 120 m to meet her there. Together, they walk a total of 425 m to get the shops. Write an equation to represent this.

The distance walked by Jennifer = d m

Use the letter requested to represent Jennifer

$$\therefore \text{Kylie walks } d + 120$$

$$\text{Jennifer} + \text{Kylie} = 425 \text{ m}$$

$$\therefore d + d + 120 \text{ m} = 425 \text{ m}$$

$$\therefore 2d + 120 \text{ m} = 425 \text{ m}$$

Collect like terms to simplify



ALWAYS finish word problems with a final statement that answers the question asked.

$$\therefore \text{The equation that represents the distance walked by Jennifer and Kylie is: } 2d + 120 \text{ m} = 425 \text{ m}$$

- (iv) Three consecutive numbers (one after the other), when added together equal 18.

Let the first number be a

Pick a letter to represent the first number

$$\therefore a, a + 1 \text{ and } a + 2 \text{ are consecutive numbers}$$

$$\therefore a + a + 1 + a + 2 = 18$$

Write the equation by adding them all together

$$\therefore 3a + 3 = 18$$

Simplify

For consecutive ODD or EVEN numbers, you add 2 instead of 1 at each step.



Word problems

1 Match each of the statements below with the correct equation.

- | | |
|---|---|
| a Seven more than a number is equal to five | ☐ $7 \times n = 5$ |
| b A number divided by seven equals five | ☐ $5 - n = 7$ |
| c Five minus a number equals seven | ☐ $\frac{7}{n} = 5$ |
| d Five added to a number equals seven | ☐ $n + 5 = 7$ |
| e The product of five and a number equals seven | ☐ $n + 7 = 5$ |
| f The difference between a number and five is seven | ☐ $n - 5 = 7$ |
| g Seven times a number equals five | ☐ $n \div 7 = 5$ |
| h The quotient of seven and a number equals five | ☐ $5n = 7$ |

2 Colour in the dot next to the expressions that match each of these statements:

- a Fiona's ipod is m months older than the one bought by her sister 4 months ago.
The age of Fiona's ipod is given by:
- ☐ $m - 4$ ☐ $m + 4$ ☐ $4 - m$ ☐ $4m$
- b Janet types 12 words per minute faster than her friend (f) when chatting on a social website.
The total words per minute Janet types compared with her friend is given by:
- ☐ $12 - 2f$ ☐ $12 - f$ ☐ $f + 12$ ☐ $f - 12$
- c Co Tin is one third the age of his father who is x years old. Co Tin's age is given by:
- ☐ $\frac{3}{x}$ ☐ $3x$ ☐ $\frac{x}{3}$ ☐ $x + 3$
- d Catherine runs m metres around a running track with friends Leif and Carol.
When Catherine lapped Leif, she was 400 m in front of him and 200 m in front of Carol.
Which expression will give the distance ran by all three at that time?
- ☐ $m - 600$ metre ☐ $m + 600$ metres ☐ $3m - 600$ metres ☐ $3m - 200$ metres
- e Kyle sold t tickets for a fund raising event. Marcus sold 25 less tickets. Together they sold a total of 38 tickets.
Hint: Ticket sales by Marcus = $t - 25$
- ☐ $25 + 2t = 28$ ☐ $38 - 2t = 25$ ☐ $2t - 25 = 38$ ☐ $t + 13 = 25$
- f Jackson is 13 cm shorter than the height of his sister (h). Together their heights are exactly 300 cm.
- ☐ $300 = 2h + 13$ ☐ $300 - 2h = 13$ ☐ $2h - 300 = 13$ ☐ $2h - 13 = 300$



Word problems

3 Use x as the variable to write a simplified equation that represents these problems then solve.

a Find three consecutive numbers that when added together equal 12.

The three consecutive numbers are: $x, x + 1, x + 2$

$$x + x + 1 + x + 2 = 12$$

$$3x + 3 = 12$$

$$3x + 3 - 3 = 12 - 3$$

$$3x = 9$$

$$3x \div 3 = 9 \div 3$$

$$x = 3$$

Simplified equation

Solve equation for x



$\therefore$ The three consecutive numbers are: $3, 3 + 1, 3 + 2$

$= 3, 4, 5$ Use solution to find the numbers

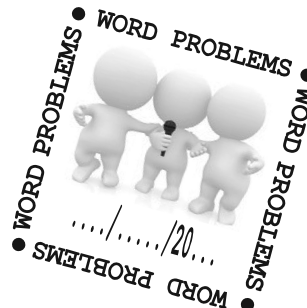
b Find three consecutive ODD numbers that when added together equal 45.

Hint: Three consecutive odd numbers are: $x, x + 2, x + 4$

c The sum of three consecutive numbers is 78. Find the three numbers.

If you use $x - 1, x$ and $x + 1$ the equation formed is simpler to solve. Try it!

d The sum of four consecutive EVEN numbers is 44. Find the four numbers.





Word problems

4 Write equations for the following problems and then solve.

- a Xian thinks of a number n . After multiplying the number by 6 and then adding 2, the answer is 26. What number is Xian thinking of?

$$\begin{aligned} 6n + 2 &= 26 \\ 6n + 2 - 2 &= 26 - 2 \\ 6n &= 24 \\ 6n \div 6 &= 24 \div 6 \\ n &= 4 \end{aligned}$$

Problem as an equation

Solve equation for n

Therefore Xian is thinking of the number 4

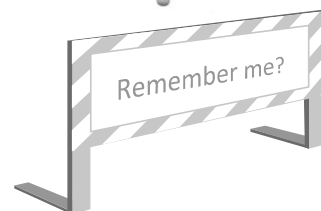


Answer with a statement

- b Freddy thinks of a number also. After multiplying the number by 2, he subtracts 11 and the result is 3 times the original number. What number is Freddy thinking of? Let the number be x .

- c Kim's pet dog weighs 4.5 kg less than the dog living next door. Together, both dogs weigh a total of 25 kg. How much does Kim's dog weigh? Hint: Let the weight of the dog next door equal d kg

- d A number plus one is multiplied by three. This is equal to seven times a number minus five. What is the value of the number? Let the number be n .



- e A number minus five is multiplied by seven and divided by two. This equals the number plus two all multiplied by three. What is the number?

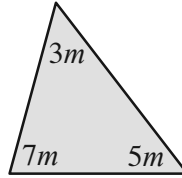
Measurement problems

Simplify and solve the following:

- (i) Find the value of m in this triangle:



$\angle$ sum = 180°



$$3m + 5m + 7m = 180^\circ$$

Angles in a triangle add up to 180°

$$15m = 180^\circ$$

Simplify sum of the angles

$$15m \div 15 = 180^\circ \div 15$$

Solve equation for m

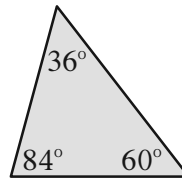
$$\therefore m = 12$$

The size of each angle can be calculated by substituting the value for m back into each angle expression.

$$\begin{aligned}\therefore 3m &= 3 \times 12 \\ &= 36\end{aligned}$$

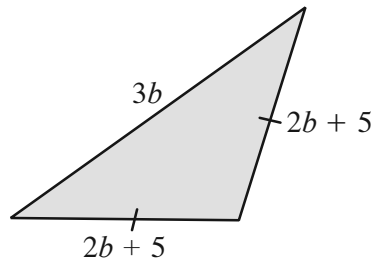
$$\begin{aligned}5m &= 5 \times 12 \\ &= 60\end{aligned}$$

$$\begin{aligned}7m &= 7 \times 12 \\ &= 84\end{aligned}$$



$$36^\circ + 60^\circ + 84^\circ = 180^\circ$$

- (ii) The perimeter of the triangle below is 73 units. Calculate the value of b .



$$3b + 2b + 5 + 2b + 5 = 73$$

Perimeter = sum of all the side lengths

$$7b + 10 = 73$$

Simplify by collecting like terms

$$7b + 10 - 10 = 73 - 10$$

Solve equation for b

$$7b = 63$$

$$7b \div 7 = 63 \div 7$$

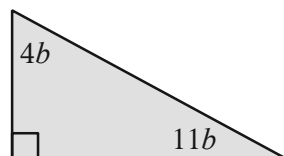
$$\therefore b = 9$$



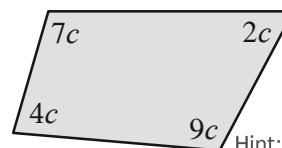
Measurement problems

1 Write an equation and calculate the value of the variable in each of the diagrams below:

a



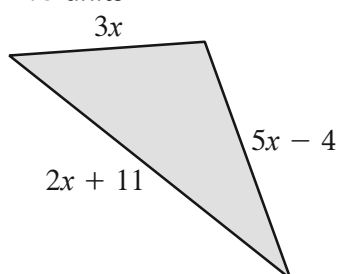
b



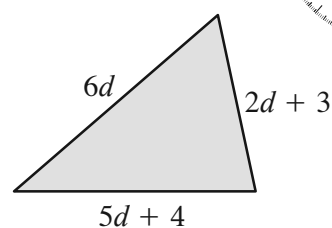
Hint: Angles add to 360°

2 Find the value of the variable in each of the diagrams below:

a Perimeter = 73 units



b Perimeter = 90 units



Formulae

Formulae show how different measurements come together for special calculations.

Use substitution to calculate the values represented by the following formulae.

- (i) The formula $V = IR$ calculates the voltage in a circuit where V is the voltage (in volts), I is the current (in amperes) and R is the resistance of the circuit (in Ohms).
Calculate the voltage V if the current I is 25 amperes and the resistance R is 12 ohms.

$$\begin{aligned} \text{When } I = 25 \text{ and } R = 12, V &= 25 \times 12 && \text{Substitute values into the formula} \\ V &= 300 \text{ Volts} \end{aligned}$$

- (ii) The length of the longest side of a right-angled triangle (c) is found using $c = \sqrt{a^2 + b^2}$ where a and b are the lengths of the other shorter sides. Calculate the length of the longest side of a right-angled triangle with short sides of length $a = 5$ cm and $b = 12$ cm.

$$\begin{aligned} \text{When } a = 5 \text{ cm and } b = 12 \text{ cm, } c &= \sqrt{5^2 + 12^2} \\ c &= \sqrt{25 + 144} \\ c &= \sqrt{169} \\ c &= 13 \end{aligned}$$

After substitution, we are sometimes left with an equation to solve.

Substitute the given values into these formulae and solve the equation for the unknown variable.

- (i) The perimeter of a rectangle (P) is calculated using the formula $P = 2l + 2b$ where l represents the length of the rectangle and b represents the breadth.
Calculate the breadth of the rectangle with a perimeter (P) of 34 cm and length (l) of 11 cm.

$$\begin{aligned} \text{When } P = 34 \text{ and } l = 11, 34 &= 2 \times 11 + 2b \text{ cm} && \text{Substitute values into the formula} \\ 34 - 22 &= 22 - 22 + 2b && \text{Solve the equation to find } b \\ 12 &= 2b \\ 12 \div 2 &= 2b \div 2 \\ 6 &= b \end{aligned}$$

$\therefore$ The breadth of the rectangle is 6 cm Answer question with a statement

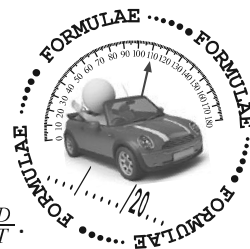
- (ii) The area of a triangle is found using the formula $A = \frac{bh}{2}$ where b represents the length of the base side and h represents the perpendicular height of the vertex opposite the base side.
What is the height of a triangle with a base length of 12 cm and area of 45 cm²?

$$\begin{aligned} \text{When } A = 45 \text{ cm and } b = 12 \text{ cm, } 45 &= \frac{12h}{2} \text{ cm}^2 && \text{Substitute values into the formula} \\ 45 \times 2 &= \frac{12h}{2} \times 2 && \text{Solve the equation to find } h \\ 90 &= 12h \\ 90 \div 12 &= 12h \div 12 \\ 7.5 &= h \end{aligned}$$

$\therefore$ The height of the triangle is 7.5 cm Answer question with a statement



Formulae



1 Calculate the value of these formulas using the given values for each variable.

- a The speed (S) of an object in kilometres per hour is given by the formula $S = \frac{D}{T}$.
 D is the distance in kilometres and T the time taken in hours. Calculate S when:

(i) $D = 140\text{ km}$ and $T = 2$ hours

(ii) $D = 31.5\text{ km}$ and $T = 0.5$ hours

$\therefore$ The speed $S = \boxed{}$ km/h

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- b To convert the temperature from degrees Fahrenheit (F) to degrees Celsius (C) the formula is:
 $C = \frac{5}{9}(F - 32)$. Calculate C when:

(i) $F = 50^\circ$

(ii) $F = 81.5^\circ$

$\therefore$ The converted temperature = $\boxed{}^\circ\text{C}$

$\therefore$ The converted temperature = $\boxed{}^\circ\text{C}$

2 Substitute the given values into these formulae and solve the equation for the unknown variable.

- a The area of a rectangle is calculated using the formula $A = lb$ where l and b are the length and breadth.
 Calculate the length l of the following rectangles with these breadths and areas:

(i) $A = 36\text{ cm}^2$ and $b = 4\text{ cm}$

(ii) $A = 25.48\text{ cm}^2$ and $b = 2.6\text{ cm}$

$\therefore$ The length l of the rectangle = $\boxed{}$ cm

$\therefore$ The length l of the rectangle = $\boxed{}$ cm

- b The average of two numbers ($\bar{x}$) is calculated by the formula $\bar{x} = \frac{(a + b)}{2}$, where a and b are
 the two numbers. Calculate the number a for following average values $\bar{x}$ and given number b .

(i) $\bar{x} = 21$ and $b = 28$

(ii) $\bar{x} = 4.75$ and $b = 3.6$

$\therefore$ The number $a = \boxed{}$

$\therefore$ The number $a = \boxed{}$



Formulae

2 Calculate the value of these formulas using the given values for each variable.

- c The speed (S) of an object in kilometres per hour is given by the formula $S = \frac{D}{T}$.
 D is the distance in kilometres and T the time taken in hours. Calculate T when:

(i) $D = 35\text{km}$ and $S = 7\text{km/h}$

(ii) $D = 300\text{km}$ and $S = 60\text{km/h}$

$\therefore$ The Time T taken = hours

$\therefore$ The Time T taken = hours

- d To convert the temperature from degrees Fahrenheit (F) to degrees Celsius (C) the formula is:
 $C = \frac{5}{9}(F - 32)$. Calculate F when:

(i) $C = 60^\circ$

(ii) $C = 27.5^\circ$

$\therefore$ The converted temperature = $^\circ F$

$\therefore$ The converted temperature = $^\circ F$

- e The formula for the area of a trapezium is: $A = \frac{h}{2}(a + b)$ where h is the distance (height) between the parallel sides and a and b are the lengths of each parallel side. Calculate the height h when:

(i) $A = 44\text{cm}^2$, $a = 4$ and $b = 7\text{cm}$

(ii) $A = 25.48\text{cm}^2$, $a = 8$ and $b = 15\text{cm}$

$\therefore$ The height h = cm

$\therefore$ The height h = cm

**Reflection Time**

Reflecting on the work covered within this booklet:

- 1 What useful skills have you gained by learning about equations?

- 2 Write about one way you think you could apply equations to a real life situations.

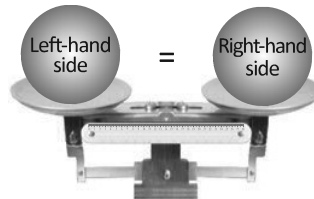
- 3 If you discovered or learnt about any shortcuts to help with equations or some other cool facts, jot them down here:



Here is a summary of the things you need to remember for Equations

Keeping things in balance

An equation is an algebraic expression where the left-hand side equals the right-hand side.



Opposite operations

Pairs of mathematical operations that do the opposite to each other.

$\oplus$ and $\ominus$ are opposite operations $\otimes$ and $\div$ are opposite operations

Solving simple equations

Solving an equation means: “What value does the variable need to be to keep the equation in balance?”

The aim is always to get the variable by itself on one side of the equation using opposite operations.

Solutions check

Always ask yourself: Does my solution ‘feel’ correct? The best way to find out is by checking your answer to see.

Equations with variables on both sides

The aim is to get all the variables on one side and the numbers on the other side of the equation.

Equations with fractions

If one fraction on either side of the equation, cross multiply the denominators.

Equations with parentheses

Two methods: 1. Divide both sides by the number out the front first.

2. Use the distributive law to expand the parentheses first.

Word problems

Change word problems into equations to help solve. Give one of the unknown values a variable.

Formulae

Use substitution to calculate the value of the formula.

Sometimes an equation remains after substitution. This can then be solved as usual.